

1 (a) Define momentum.

[1]

- (b) A ball X of mass 0.15 kg moves along a horizontal frictionless surface. The ball collides elastically with ball Y that is initially stationary. Balls X and Y have equal diameters but different masses.

Figure 1.1 shows the variation with time of the velocity of ball X before, during and after the collision.

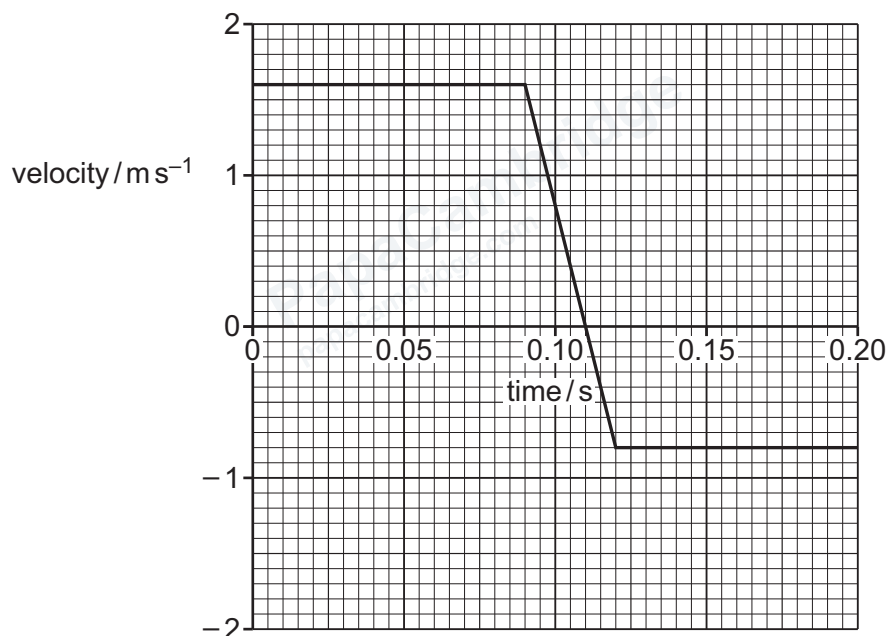


Figure 1.1

- (i) Determine the magnitude of the acceleration of ball X during the collision.

acceleration = m s^{-2} [2]

- (ii) Determine the magnitude of the change in momentum of ball X due to the collision.

change in momentum = kg m s^{-1} [2]

- (iii) Determine the total distance travelled by ball X during the collision.

distance = m [2]

- (c) (i) By considering relative speeds, explain why the velocity of ball Y after the collision must be $+0.80 \text{ m s}^{-1}$.

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.....
..... [2]

- (ii) On Figure 1.1, sketch the variation of the velocity of ball Y with time between time = 0 and time = 0.20 s. [2]

- (iii) Calculate the mass of ball Y.

mass = kg [2]

[Total: 13]